

Norway

SLEIPNER

Volve F

F-4

F-4

Plan: F4 Rev 0.16 FinalCoords CC 310707

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table (As Installed) @ 54.90m (As Installed)
Site:	Volve F	MD Reference:	Rotary Table (As Installed) @ 54.90m (As Installed)
Well:	F-4	North Reference:	Grid
Wellbore:	F-4	Survey Calculation Method:	Minimum Curvature
Design:	F4 Rev 0.16 FinalCoords CC 310707	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-4			
Well Position	+N/-S	-2.70 m	Northing:	6,478,560.83 m
	+E/-W	-0.19 m	Easting:	435,049.83 m
Position Uncertainty	0.00 m	Wellhead Depth:	91.00 m	Water Depth: 91.00 m

Wellbore	F-4				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	NETICREFERENCE	19.10.2005	-2.55	71.62	50,281

Design	F4 Rev 0.16 FinalCoords CC 310707			
Audit Notes:				
Version:	Phase:	PLAN	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	-2.70	-0.19	150.36

Survey Tool Program	Date	31.05.2010		
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
145.90	500.90	F4 Rev 0.16 FinalCoords CC 310707 (F-4)	Wellbore Surveyor, stat	Gyro Tool from GD
500.90	3,631.82	F4 Rev 0.16 FinalCoords CC 310707 (F-4)	Magnetic, std, mag-corr	Magnetic Tools (MWD, EMS)

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-2.70	-0.19	0.00	0.000	0.000	0.000	
284.90	0.00	0.00	284.90	-2.70	-0.19	0.00	0.000	0.000	0.000	
524.90	8.00	220.00	524.12	-15.51	-10.94	5.82	1.000	1.000	0.000	
905.43	13.00	150.00	899.46	-73.10	-6.54	58.04	1.000	0.394	-5.519	
1,317.43	13.00	150.00	1,300.90	-153.36	39.80	150.72	0.000	0.000	0.000	
2,525.97	38.03	102.20	2,391.74	-354.56	482.21	544.40	0.760	0.621	-1.187	
2,575.97	38.03	102.20	2,431.12	-361.07	512.31	564.95	0.000	0.000	0.000	
3,312.77	55.00	200.00	2,999.89	-737.09	646.85	958.30	2.747	0.691	3.982	

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Planned Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
3,631.82	55.00	200.00	3,182.89	-982.68	557.46	1,127.54	0.000	0.000	0.000

Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
F4 Alternative 201106 - plan misses target center by 202.02m at 3312.77m MD (2999.89 TVD, -737.09 N, 646.85 E) - Point	0.00	0.00	2,984.90	-913.84	550.17	6,477,650.00	435,600.00	58° 26' 0.5694 N	1° 53' 49.7589 E
Volve F4 201006 - plan misses target center by 0.41m at 3312.67m MD (2999.83 TVD, -737.02 N, 646.88 E) - Point	0.00	0.00	2,999.90	-736.78	647.20	6,477,827.00	435,697.00	58° 26' 6.3425 N	1° 53' 55.5590 E
F4 50mSW of v11 - plan misses target center by 49.58m at 3312.77m MD (2999.89 TVD, -737.09 N, 646.85 E) - Point	0.00	0.00	2,999.90	-783.76	630.11	6,477,780.04	435,679.91	58° 26' 4.8153 N	1° 53' 54.5530 E
F4 100m SW of v11 - plan misses target center by 99.58m at 3312.77m MD (2999.89 TVD, -737.09 N, 646.85 E) - Point	0.00	0.00	2,999.90	-830.74	613.01	6,477,733.08	435,662.82	58° 26' 3.2881 N	1° 53' 53.5468 E
Volve I20 - plan misses target center by 34.08m at 3312.77m MD (2999.89 TVD, -737.09 N, 646.85 E) - Point	0.00	0.94	3,000.90	-765.79	665.21	6,477,798.00	435,715.00	58° 26' 5.4145 N	1° 53' 56.6978 E
Volve I21 Bunn - plan misses target center by 136.34m at 3312.77m MD (2999.89 TVD, -737.09 N, 646.85 E) - Point	0.00	0.94	3,131.90	-765.79	665.21	6,477,798.00	435,715.00	58° 26' 5.4145 N	1° 53' 56.6978 E

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
250.90	250.90	30"	30.000	36.000	
1,350.90	1,333.48	13 3/8"	13.375	17.500	
2,804.90	2,619.63	9 5/8"	9.625	12.250	
3,536.90	3,128.45	7"	7.000	8.500	

Checked By: _____	Approved By: _____	Date: _____
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